

SHIVARATHRI KARTHIK

AI/ML Engineer | Generative AI | Agentic AI Engineer

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PROFESSIONAL SUMMARY

AI/ML Engineer with nearly 4 years of experience in designing, developing, and deploying enterprise AI applications across Machine Learning, Data Science, Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and Agentic AI. Experienced in delivering end-to-end AI applications for enterprise clients in the automotive and banking domains through IBM, with expertise spanning business requirement analysis, data engineering, model development, LLM integration, deployment, monitoring, and production support.

Specialized in architecting and developing AI Agents and multi-agent AI systems using LangGraph, LangChain, CrewAI, AutoGen, and Agentic AI frameworks, enabling autonomous reasoning, planning, tool calling, memory management, and workflow orchestration. Hands-on experience building enterprise RAG applications using Azure OpenAI, GPT-4o, Claude, Gemini, Vector Databases including FAISS, Pinecone, Weaviate, and Azure AI Search, integrating enterprise data from SAP ECC/SAP S/4HANA, SharePoint, Confluence, Jira, SQL Server, REST APIs, and document repositories for AI/ML and Generative AI solutions.

Strong expertise in Machine Learning, Deep Learning, NLP, predictive analytics, anomaly detection, demand forecasting, classification, regression, recommendation systems, and time-series forecasting using Python, Scikit-learn, XGBoost, TensorFlow, and PyTorch. Experienced in developing cloud-native AI applications using Python, FastAPI, Docker, Kubernetes, Azure, AWS, GitHub Actions, Azure DevOps, and MLflow, implementing scalable LLMops, MLOps, CI/CD pipelines, model monitoring, prompt engineering, Responsible AI, and production-grade AI solutions. Passionate about building intelligent, scalable, and business-focused AI systems while collaborating with cross-functional teams using Agile methodologies.

- **Nearly 4 years of experience designing, developing, deploying, and supporting enterprise AI applications as an AI/ML Engineer, Data Scientist, GenAI Engineer, and Agentic AI Engineer.**
- Designed and deployed **Agentic AI and Multi-Agent Systems** using **LangGraph, LangChain, CrewAI, AutoGen, ReAct, and tool-calling architectures.**
- Built enterprise **Retrieval-Augmented Generation (RAG)** applications using **Azure OpenAI, GPT-4o, Claude, Gemini, FAISS, Pinecone, Weaviate, and Azure AI Search.**
- Developed end-to-end **Machine Learning and Data Science** solutions including predictive analytics, demand forecasting, anomaly detection, recommendation systems, regression, classification, NLP, and time-series forecasting.
- Experienced in the complete AI lifecycle, including **requirement gathering, data collection, ETL, feature engineering, model development, evaluation, deployment, monitoring, optimization, and production support.**
- Built scalable REST APIs and microservices using **Python, FastAPI**, integrating enterprise applications, databases, cloud services, and external APIs.
- Hands-on experience with **Azure, AWS, Docker, Kubernetes, GitHub Actions, Azure DevOps, MLflow, LLMops, MLOps, CI/CD, and cloud-native AI deployment.**
- Strong knowledge of **Prompt Engineering, Prompt Optimization, Function Calling, Tool Calling, Vector Search, Semantic Search, Hybrid Search, Embedding Models, Memory Management, and AI Governance.**
- Experienced working with **cross-functional teams, business stakeholders, solution architects, and Agile/Scrum methodologies** to deliver scalable, production-ready AI solutions.
- Delivered enterprise AI solutions across automotive and banking domains using Machine Learning, Generative AI, RAG, and Agentic AI.

Technical Skills:

- **Programming Languages:** Python, SQL.
- **Agentic AI & Generative AI:** Agentic AI, AI Agents, Multi-Agent Systems, LangGraph, LangChain, CrewAI, AutoGen, Retrieval-Augmented Generation (RAG), Prompt Engineering, Prompt Optimization, Function Calling, Tool Calling, AI Workflow Orchestration, Memory Management.
- **Large Language Models (LLMs):** GPT-4o, GPT-4, Azure OpenAI, Claude, Gemini, Llama, Hugging Face Transformers.
- **Machine Learning & Data Science:** Regression, Classification, Clustering, Time Series Forecasting (ARIMA, Prophet), XGBoost, LightGBM, Random Forest, Decision Trees, Logistic Regression, Feature Engineering, Feature Selection, Hyperparameter Tuning, Anomaly Detection, Model Evaluation.
- **Deep Learning & NLP:** TensorFlow, PyTorch, Keras, ANN, CNN, RNN, LSTM, GRU, Transformers, NLTK, SpaCy.
- **Vector Databases & Semantic Search:** FAISS, Pinecone, Weaviate, Azure AI Search, OpenSearch, Vector Search, Hybrid Search, Semantic Search, Embedding Models.
- **Backend Development:** FastAPI, Flask, REST APIs, Microservices.
- **Data Engineering:** Pandas, NumPy, PySpark, Apache Kafka, ETL Pipelines, Data Lakes, Data Warehousing.
- **LLMOps, MLOps & CI/CD:** MLflow, Kubeflow, GitHub Actions, Azure DevOps, AWS CodePipeline, CI/CD Pipelines, Model Deployment, Model Monitoring.
- **Cloud & DevOps:** Microsoft Azure, Azure OpenAI, AWS, Amazon Bedrock, AWS SageMaker, Docker, Kubernetes.
- **Enterprise Applications & ERP:** SAP ECC, SAP S/4HANA, SAP HANA, SAP OData Services, SharePoint, Confluence, Jira
- **Databases:** SQL Server, PostgreSQL, MySQL.
- **Visualization & Development Tools:** Power BI, Tableau, Git, GitHub, Jupyter Notebook, VS Code, Jira, Agile/Scrum.

PROFESSIONAL EXPERIENCE

Senior AI/ML Engineer | Generative AI | Agentic AI Engineer IBM | Aug 2022 – Present

Working as a Senior AI/ML Engineer with nearly 4 years of experience in designing, developing, and deploying enterprise AI applications across Machine Learning, Data Science, Generative AI, Retrieval-Augmented Generation (RAG), and Agentic AI. Experienced in building production-ready AI Agents, enterprise Large Language Model (LLM) solutions, intelligent document processing systems, predictive analytics platforms, and cloud-native AI applications using Python, Azure OpenAI, LangGraph, LangChain, FastAPI, Docker, Kubernetes, Azure, and AWS. Responsible for the complete AI lifecycle, including business requirement gathering, solution architecture, data engineering, feature engineering, model development, deployment, LLMOps, MLOps, CI/CD automation, monitoring, optimization, production support, and stakeholder collaboration while following Agile/Scrum methodologies.

Client: Citizens Bank, Rhode Island Senior Agentic AI Engineer

Duration: Jul 2025 – Present

Project Overview:

Designed and developed an enterprise Agentic AI platform that automates intelligent document processing, enterprise knowledge retrieval, workflow orchestration, and AI-driven decision support using multi-agent architectures and Large Language Models (LLMs).

Responsibilities

- Developed and optimized Deep Learning models using TensorFlow, Keras, and PyTorch for intelligent document classification, information extraction, and AI-driven decision support.
- Designed and fine-tuned neural network architectures including ANN, CNN, and LSTM using Python to improve model accuracy, inference performance, and enterprise AI workflows.
- Built scalable Python-based Deep Learning pipelines for data preprocessing, model training, validation, and inference, following production-ready ML engineering best practices.
- Collaborated with business stakeholders, solution architects, and SMEs to gather business and technical requirements for enterprise AI automation.
- Architected and implemented enterprise AI Agent solutions using LangGraph, LangChain, CrewAI, and AutoGen.
- Developed Planner, Retrieval, Research, Validation, Reflection, Memory, and Tool Execution Agents for autonomous workflow orchestration.
- Built enterprise-grade Retrieval-Augmented Generation (RAG) pipelines integrating Azure OpenAI, GPT-4o, Claude, Gemini, and Llama.
- Integrated enterprise SAP data (SAP ECC/S/4HANA) into RAG pipelines using APIs and ETL processes for AI-powered knowledge retrieval.
- Implemented OCR, semantic chunking, metadata extraction, embedding generation, vector search, hybrid search, and reranking techniques.
- Designed prompt engineering, prompt optimization, function calling, structured outputs, AI guardrails, and tool-calling workflows to improve LLM response quality.
- Integrated AI Agents with enterprise applications, SQL databases, REST APIs, and third-party business systems.
- Developed scalable backend microservices and REST APIs using Python and FastAPI.
- Implemented conversation memory, vector memory, long-term memory, authentication, authorization, and RBAC.
- Deployed AI applications using Docker, Kubernetes, Azure OpenAI, and Azure AI Search.
- Automated CI/CD pipelines using Azure DevOps and GitHub Actions.
- Implemented LLM Ops practices including prompt versioning, model evaluation, monitoring, token optimization, and response quality assessment.
- Collaborated with cross-functional teams throughout development, testing, deployment, production support, and technical documentation.
- Participated in Agile Scrum ceremonies, sprint planning, release management, and code reviews.

Key Achievements

- Improved enterprise knowledge retrieval using semantic search and AI Agent orchestration.
- Reduced manual effort by automating intelligent document retrieval and enterprise workflows.
- Enhanced response quality through optimized RAG pipelines, prompt engineering, and LLM evaluation techniques.

Technologies:

Python | LangGraph | LangChain | CrewAI | AutoGen | Azure OpenAI | GPT-4o | Claude | Gemini | Llama | FastAPI | FAISS | Pinecone | Azure AI Search | Docker | Kubernetes | Azure DevOps | GitHub Actions | MLflow

Client: Ford Motor Company
GenAI Engineer

Duration: Jan 2024 – Jun 2025

Project Overview:

Designed and developed enterprise Generative AI applications for intelligent knowledge management, document understanding, contextual question answering, and enterprise search using Retrieval-Augmented Generation (RAG).

Responsibilities

- Gathered business requirements and identified enterprise Generative AI use cases.
- Designed and developed enterprise RAG applications using Azure OpenAI and Large Language Models (LLMs).
- Built document ingestion pipelines Retrieved and integrated enterprise data from SAP systems for contextual search and Generative AI applications.
- Performed OCR, metadata extraction, semantic chunking, embedding generation, and intelligent document indexing.
- Developed semantic search solutions using FAISS, Pinecone, and Azure AI Search.
- Integrated Azure OpenAI GPT models for contextual question answering, summarization, and intelligent knowledge retrieval.
- Developed prompt engineering, prompt optimization, citation-aware responses, hybrid search, and reranking techniques.
- Developed REST APIs using FastAPI and integrated AI services with enterprise applications.
- Implemented Azure Active Directory authentication and role-based access control.
- Optimized retrieval accuracy, embedding quality, and response latency.
- Automated deployments using Docker, Kubernetes, Azure DevOps, and GitHub Actions.
- Implemented LLM monitoring, prompt evaluation, MLflow tracking, and production support.
- Collaborated with business users, developers, QA teams, and solution architects throughout the software development lifecycle.

Key Achievements

- Improved enterprise document search accuracy through optimized RAG architecture.
- Reduced document retrieval time using semantic search and vector databases.
- Improved AI response quality through prompt optimization and hybrid search strategies.

Technologies:

Python | Azure OpenAI | GPT-4 | LangChain | FastAPI | FAISS | Pinecone | Azure AI Search | Docker | Kubernetes | Azure DevOps | GitHub Actions | MLflow | Azure SQL

Client: General Motors (US Remote)

AI/ML Engineer – Data Scientist

Duration: Aug 2022 – Dec 2023

Project Overview:

Designed and deployed enterprise Machine Learning solutions for predictive analytics, demand forecasting, anomaly detection, and business intelligence using scalable cloud-native ML architectures.

Responsibilities

- Collaborated with business stakeholders to understand predictive analytics, forecasting, and business intelligence requirements.
- Collected structured and unstructured data from SAP ECC/SAP S/4HANA, SQL Server, Oracle, enterprise databases, REST APIs, and transactional systems.
- Performed ETL, data preprocessing, and feature engineering on SAP enterprise data for predictive analytics and machine learning models.
- Performed exploratory data analysis (EDA), data cleaning, transformation, feature engineering, and data validation.
- Designed, trained, and deployed machine learning models for predictive analytics, demand forecasting, anomaly detection, and customer analytics.
- Implemented Random Forest, XGBoost, Gradient Boosting, Decision Trees, Logistic Regression, and K-Means clustering algorithms.
- Performed hyperparameter tuning using Grid Search, Random Search, and cross-validation.
- Evaluated model performance using Accuracy, Precision, Recall, F1-Score, ROC-AUC, RMSE, MAE, and

MAPE.

- Implemented automated model training, experiment tracking, retraining, and deployment pipelines using MLflow.
- Developed REST APIs using FastAPI and Flask for enterprise ML model deployment.
- Deployed machine learning applications using Docker and Microsoft Azure.
- Developed Power BI dashboards and interactive reports for business users and executive stakeholders.
- Automated CI/CD pipelines using GitHub Actions and collaborated with DevOps teams for production deployments.
- Participated in Agile ceremonies, production support, sprint planning, and technical documentation.
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Key Achievements

- Improved predictive model performance through feature engineering and hyperparameter optimization.
- Automated ML deployment and monitoring pipelines using MLflow and CI/CD.
- Delivered scalable analytics solutions supporting enterprise decision-making and operational efficiency.

Technologies:

Python | Scikit-learn | Pandas | NumPy | XGBoost | Random Forest | SQL | FastAPI | Flask | MLflow | Docker | Microsoft Azure | Power BI | GitHub Actions

EDUCATION:

- **Bachelor of Technology in Computer Science and Engineering from JNTUH 2022.**

CERTIFICATIONS:

- Microsoft Azure AI Fundamentals
- AWS Machine Learning
- IBM Generative AI
- DeepLearning.AI Generative AI